

FIGURE 7.3.13 Molecule in Problem 49

Discussion Problems

50. Use the dot product to prove the Cauchy-Schwarz inequality: $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$.
51. Use the dot product to prove the triangle inequality $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$. [Hint: Consider property (vi) of the dot product.]
52. Prove that the vector $\mathbf{n} = a\mathbf{i} + b\mathbf{j}$ is perpendicular to the line whose equation is $ax + by + c = 0$. [Hint: Let $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ be distinct points on the line.]

53. Use the result of Problem 52 and FIGURE 7.3.14 to show that the distance d from a point $P_1(x_1, y_1)$ to a line $ax + by + c = 0$ is

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}.$$

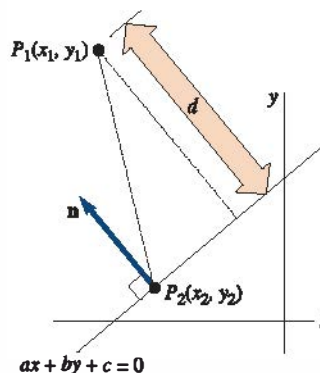


FIGURE 7.3.14 Distance d in Problem 53

7.4 Cross Product

Introduction The dot product, introduced in the preceding section, works in both two- and three-dimensional spaces and results in a *number*. On the other hand, the **cross product**, introduced in this section, is only defined for vectors in 3-space and results in another *vector* in 3-space.

Before proceeding we need the following two results from the theory of determinants.

Review of Determinants A determinant of order 2 is the number

$$\begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1. \quad (1)$$

A determinant of order 3 is the number defined in terms of three determinants of order 2:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}. \quad (2)$$

This is called **expanding the determinant** along the first row. For example, from (1)

$$\begin{vmatrix} -4 & -2 \\ 5 & 3 \end{vmatrix} = (-4)3 - (-2)5 = -2$$

and from (2)

$$\begin{vmatrix} 8 & 5 & 4 \\ 2 & 4 & 6 \\ -1 & 2 & 3 \end{vmatrix} = 8 \begin{vmatrix} 4 & 6 \\ 2 & 3 \end{vmatrix} - 5 \begin{vmatrix} 2 & 6 \\ -1 & 3 \end{vmatrix} + 4 \begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = 8(0) - 5(12) + 4(8) = -28.$$

Component Form of the Cross Product As we did in the discussion of the dot product, we define the **cross product** of two vectors $\mathbf{a}$ and $\mathbf{b}$ in terms of the components of the vectors.

Definition 7.4.1 Cross Product of Two Vectors

The **cross product** of two vectors $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is the vector

$$\mathbf{a} \times \mathbf{b} = (a_2 b_3 - a_3 b_2)\mathbf{i} - (a_1 b_3 - a_3 b_1)\mathbf{j} + (a_1 b_2 - a_2 b_1)\mathbf{k}. \quad (3)$$

See Sections 8.4 and 8.5 for a complete discussion of determinants.

The coefficients of the basis vectors in (3) are recognized as determinants of order 2, so (3) can be written as

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k}. \quad (4)$$

By comparing (4) with (2), we see that the cross product can be formally expressed as a determinant of order 3 with the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ as entries in the first row and the components of $\mathbf{a}$ and $\mathbf{b}$ entries of the second and third rows, respectively:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}. \quad (5)$$

EXAMPLE 1 Cross Product Using (5)

Let $\mathbf{a} = 4\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$. Find $\mathbf{a} \times \mathbf{b}$.

SOLUTION We use (5) and expand the determinant along the first row:

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & -2 & 5 \\ 3 & 1 & -1 \end{vmatrix} = \begin{vmatrix} -2 & 5 \\ 1 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 4 & 5 \\ 3 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 4 & -2 \\ 3 & 1 \end{vmatrix} \mathbf{k} \\ &= -3\mathbf{i} + 19\mathbf{j} + 10\mathbf{k}. \end{aligned}$$

EXAMPLE 2 Cross Product of Two Basis Vectors

If $\mathbf{i} = \langle 1, 0, 0 \rangle$ and $\mathbf{j} = \langle 0, 1, 0 \rangle$, then (5) gives

$$\mathbf{i} \times \mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \mathbf{k} = \mathbf{k}$$

$$\text{and} \quad \mathbf{i} \times \mathbf{i} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} \mathbf{k} = \mathbf{0}.$$

Proceeding as in Example 2, it is readily shown that

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \quad (6)$$

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}, \quad \mathbf{k} \times \mathbf{j} = -\mathbf{i}, \quad \mathbf{i} \times \mathbf{k} = -\mathbf{j}, \quad (7)$$

$$\text{and} \quad \mathbf{i} \times \mathbf{i} = \mathbf{0}, \quad \mathbf{j} \times \mathbf{j} = \mathbf{0}, \quad \mathbf{k} \times \mathbf{k} = \mathbf{0}. \quad (8)$$

The results in (6) can be obtained using the circular mnemonic illustrated in **FIGURE 7.4.1**. Notice in (7), if we cross two basis vectors in the direction opposite to that shown in Figure 7.4.1, then we get the negative of the corresponding vector in (6). The results in (6), (7), and (8) are special cases of (ii) and (vi) in Theorem 7.4.1.

Properties The next theorem summarizes some of the important properties of the cross product.

Theorem 7.4.1 Properties of the Cross Product

- (i) $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ if $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$
- (ii) $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$
- (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c})$ ← distributive law
- (iv) $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \times \mathbf{c}) + (\mathbf{b} \times \mathbf{c})$ ← distributive law
- (v) $\mathbf{a} \times (k\mathbf{b}) = (k\mathbf{a}) \times \mathbf{b} = k(\mathbf{a} \times \mathbf{b})$, k a scalar
- (vi) $\mathbf{a} \times \mathbf{a} = \mathbf{0}$
- (vii) $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$
- (viii) $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$

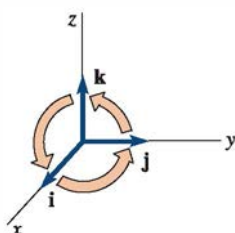


FIGURE 7.4.1 A mnemonic for cross products involving $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$

PROOF: All the properties can be proved directly from (4) and (5). To prove part (i), we let $\mathbf{a} = \langle 0, 0, 0 \rangle = \mathbf{0}$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$. Then from (5)

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 0 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 0 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & 0 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} \\ &= [0(b_3) - 0(b_2)]\mathbf{i} - [0(b_3) - 0(b_1)]\mathbf{j} + [0(b_2) - 0(b_1)]\mathbf{k} \\ &= 0\mathbf{i} + 0\mathbf{j} + 0\mathbf{k} = \mathbf{0}.\end{aligned}$$

To prove part (iii) we let $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, and $\mathbf{c} = \langle c_1, c_2, c_3 \rangle$. Then

$$\begin{aligned}\mathbf{a} \times (\mathbf{b} + \mathbf{c}) &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 + c_1 & b_2 + c_2 & b_3 + c_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 + c_1 & b_3 + c_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 + c_1 & b_2 + c_2 \end{vmatrix} \mathbf{k} \\ &= [(a_2b_3 + a_2c_3) - (a_3b_2 + a_3c_2)]\mathbf{i} - [(a_1b_3 + a_1c_3) - (a_3b_1 + a_3c_1)]\mathbf{j} \\ &\quad + [(a_1b_2 + a_1c_2) - (a_2b_1 + a_2c_1)]\mathbf{k} \\ &= (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \\ &\quad + (a_2c_3 - a_3c_2)\mathbf{i} - (a_1c_3 - a_3c_1)\mathbf{j} + (a_1c_2 - a_2c_1)\mathbf{k} \\ &= (\mathbf{a} \times \mathbf{b}) + (\mathbf{a} \times \mathbf{c}).\end{aligned}$$

Because part (ii) of Theorem 7.4.1 indicates that the cross product is not commutative, there are two distributive laws in parts (iii) and (iv) of the theorem.

Parts (vii) and (viii) of Theorem 7.4.1 deserve special attention. In view of Theorem 7.3.3 we see (vii) and (viii) imply that the vector $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$ and that $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{b}$. In the case when $\mathbf{a}$ and $\mathbf{b}$ are nonzero vectors, then $\mathbf{a} \times \mathbf{b}$ is orthogonal to *every* vector in the plane containing $\mathbf{a}$ and $\mathbf{b}$. Put another way,

$$\mathbf{a} \times \mathbf{b} \text{ is orthogonal to the plane containing } \mathbf{a} \text{ and } \mathbf{b}. \quad (9)$$

You can see in Figure 7.4.1 and from the results in (6) that $\mathbf{i} \times \mathbf{j}$ is orthogonal to the plane of $\mathbf{i}$ and $\mathbf{j}$; that is, the xy -plane, $\mathbf{j} \times \mathbf{k}$ is orthogonal to the yz -plane, and $\mathbf{k} \times \mathbf{i}$ is orthogonal to the xz -plane.

The results in (6) and (7) give us a clue as to the direction in which the vector $\mathbf{a} \times \mathbf{b}$ points.

Right-Hand Rule The vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{a} \times \mathbf{b}$ form a **right-handed system** or a **right-handed triple**. This means that $\mathbf{a} \times \mathbf{b}$ points in the direction given by the **right-hand rule**:

If the fingers of the right hand point along the vector $\mathbf{a}$ and then curl toward the vector $\mathbf{b}$, the thumb will give the direction of $\mathbf{a} \times \mathbf{b}$. (10)

See **FIGURE 7.4.2(a)**. In Figure 7.4.2(b), the right-hand rule shows the direction of $\mathbf{b} \times \mathbf{a}$.

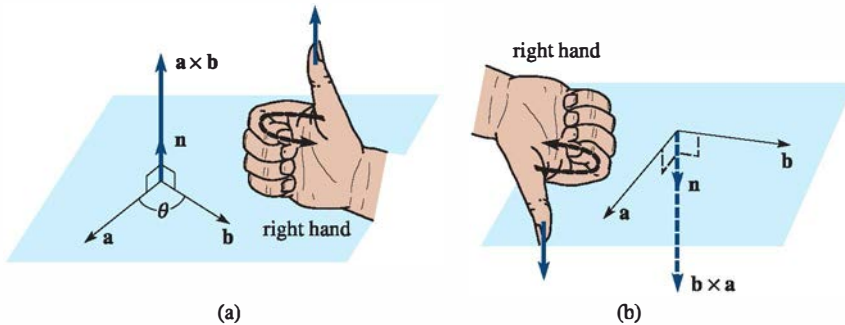


FIGURE 7.4.2 Right-hand rule

Because $\mathbf{a} \times \mathbf{b}$ is a vector its magnitude $\|\mathbf{a} \times \mathbf{b}\|$ can be found from (vii) of Definition 7.2.1. As we see in the next theorem, $\|\mathbf{a} \times \mathbf{b}\|$ can also be expressed in terms of the angle θ between the vectors $\mathbf{a}$ and $\mathbf{b}$.

Theorem 7.4.2 Magnitude of the Cross Product

For nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, if θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ ($0 \leq \theta \leq \pi$), then

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta. \quad (11)$$

PROOF: We separately compute the squares of the left- and right-hand sides of equation (11) using the component forms of $\mathbf{a}$ and $\mathbf{b}$:

$$\begin{aligned} \|\mathbf{a} \times \mathbf{b}\|^2 &= (a_2b_3 - a_3b_2)^2 + (a_1b_3 - a_3b_1)^2 + (a_1b_2 - a_2b_1)^2 \\ &= a_2^2b_3^2 - 2a_2b_3a_3b_2 + a_3^2b_2^2 + a_1^2b_3^2 - 2a_1b_3a_3b_1 + a_3^2b_1^2 \\ &\quad + a_1^2b_2^2 - 2a_1b_2a_2b_1 + a_2^2b_1^2, \\ (\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta)^2 &= \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \sin^2 \theta = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 (1 - \cos^2 \theta) \\ &= \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 - \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 \cos^2 \theta = \|\mathbf{a}\|^2 \|\mathbf{b}\|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \\ &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1b_1 + a_2b_2 + a_3b_3)^2 \\ &= a_2^2b_3^2 - 2a_2b_2a_3b_3 + a_3^2b_2^2 + a_1^2b_3^2 - 2a_1b_1a_3b_3 + a_3^2b_1^2 \\ &\quad + a_1^2b_2^2 - 2a_1b_1a_2b_2 + a_2^2b_1^2. \end{aligned}$$

Since both sides are equal to the same quantity, they must be equal to each other, so

$$\|\mathbf{a} \times \mathbf{b}\|^2 = (\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta)^2.$$

Finally, taking the square root of both sides and using the fact that $\sqrt{\sin^2 \theta} = \sin \theta$ since $\sin \theta \geq 0$ for $0 \leq \theta \leq \pi$, we have $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$. $\equiv$

Alternative Form Combining (9), (10), and Theorem 7.4.2 we see for *any* pair of vectors $\mathbf{a}$ and $\mathbf{b}$ in R^3 that the cross product has the alternative form

$$\mathbf{a} \times \mathbf{b} = (\|\mathbf{a}\| \|\mathbf{b}\| \sin \theta) \mathbf{n}, \quad (12)$$

where $\mathbf{n}$ is a unit vector given by the right-hand rule that is orthogonal to the plane of $\mathbf{a}$ and $\mathbf{b}$.

Parallel Vectors We saw in Section 7.1 that two nonzero vectors are parallel if and only if one is a nonzero scalar multiple of the other. Thus, two vectors are parallel if they have the forms $\mathbf{a}$ and $k\mathbf{a}$, where $\mathbf{a}$ is any vector and k is a scalar. By properties (v) and (vi) in Theorem 7.4.1, the cross product of parallel vectors must be $\mathbf{0}$. This fact is summarized in the next theorem.

Theorem 7.4.3 Criterion for Parallel Vectors

Two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.

Of course, Theorem 7.4.3 follows as well from (12) because the angle between parallel vectors $\mathbf{a}$ and $\mathbf{b}$ is either $\theta = 0$ or $\theta = \pi$.

EXAMPLE 3 Parallel Vectors

Determine whether $\mathbf{a} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{b} = -6\mathbf{i} - 3\mathbf{j} + 3\mathbf{k}$ are parallel vectors.

SOLUTION From the cross product

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & -1 \\ -6 & -3 & 3 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ -3 & 3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 2 & -1 \\ -6 & 3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 2 & 1 \\ -6 & -3 \end{vmatrix} \mathbf{k} \\ &= 0\mathbf{i} - 0\mathbf{j} + 0\mathbf{k} = \mathbf{0} \end{aligned}$$

and Theorem 7.4.3 we conclude that $\mathbf{a}$ and $\mathbf{b}$ are parallel vectors. $\equiv$

This more geometric form is generally used as the definition of the cross product in a physics course.

Special Products The **scalar triple product** of vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. Using the component forms of the definitions of the dot and cross products, we have

$$\begin{aligned}\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \cdot \left[\begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix} \mathbf{k} \right] \\ &= a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.\end{aligned}$$

Thus, we see that the scalar triple product can be written as a determinant of order 3:

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}. \quad (13)$$

Using (4), (5), and (2) of Section 7.3 it can be shown that

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}. \quad (14)$$

See Problem 61 in Exercises 7.4.

The **vector triple product** of three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}).$$

The vector triple product is related to the dot product by

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}. \quad (15)$$

See Problem 59 in Exercises 7.4.

Areas Two nonzero and nonparallel vectors $\mathbf{a}$ and $\mathbf{b}$ can be considered to be the sides of a parallelogram. The **area A of a parallelogram** is

$$A = (\text{base})(\text{altitude}).$$

From **FIGURE 7.4.3(a)** we see that $A = \|\mathbf{b}\|(\|\mathbf{a}\| \sin \theta) = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$

$$\text{or} \quad A = \|\mathbf{a} \times \mathbf{b}\|. \quad (16)$$

Likewise from Figure 7.4.3(b), we see that the **area of a triangle** with sides $\mathbf{a}$ and $\mathbf{b}$ is

$$A = \frac{1}{2} \|\mathbf{a} \times \mathbf{b}\|. \quad (17)$$

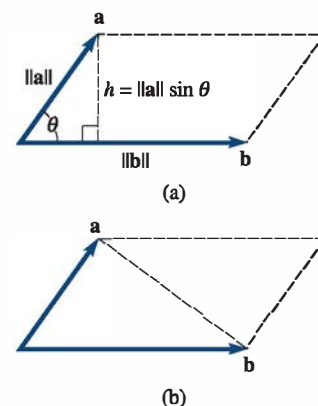


FIGURE 7.4.3 Areas of a parallelogram and a triangle

EXAMPLE 4 Area of a Triangle

Find the area of the triangle determined by the points $P_1(1, 1, 1)$, $P_2(2, 3, 4)$, and $P_3(3, 0, -1)$.

SOLUTION The vectors $\overrightarrow{P_1P_2}$ and $\overrightarrow{P_1P_3}$ can be taken as two sides of the triangle. Since $\overrightarrow{P_1P_2} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\overrightarrow{P_1P_3} = \mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$, we have

$$\begin{aligned}\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ 1 & -3 & -5 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ -3 & -5 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 3 \\ 1 & -5 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & 2 \\ 1 & -3 \end{vmatrix} \mathbf{k} \\ &= -\mathbf{i} + 8\mathbf{j} - 5\mathbf{k}.\end{aligned}$$

From (17) we see that the area is

$$A = \frac{1}{2} \|\mathbf{i} - 8\mathbf{j} + 5\mathbf{k}\| = \frac{1}{2} \sqrt{1 + 64 + 25} = \frac{1}{2} \sqrt{90} = \frac{3\sqrt{10}}{2}. \quad \equiv$$

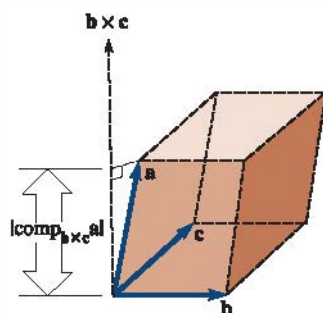


FIGURE 7.4.4 Parallelepiped formed by three vectors

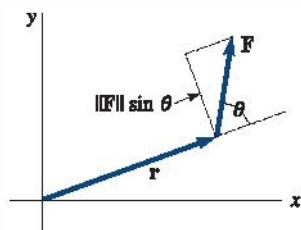


FIGURE 7.4.5 A force acting at the end of a vector

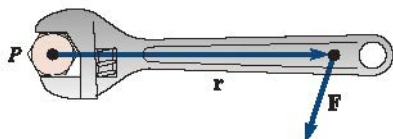


FIGURE 7.4.6 A wrench applying torque to a bolt

Volume of a Parallelepiped If the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ do not lie in the same plane, then the **volume of the parallelepiped** with edges $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ shown in **FIGURE 7.4.4** is

$$\begin{aligned} V &= (\text{area of base})(\text{height}) \\ &= \|\mathbf{b} \times \mathbf{c}\| |\text{comp}_{\mathbf{b} \times \mathbf{c}} \mathbf{a}| \\ &= \|\mathbf{b} \times \mathbf{c}\| \left| \mathbf{a} \cdot \left(\frac{1}{\|\mathbf{b} \times \mathbf{c}\|} \mathbf{b} \times \mathbf{c} \right) \right| \end{aligned}$$

or

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|. \quad (18)$$

Thus, the volume of a parallelepiped determined by three vectors is the absolute value of the scalar triple product of the vectors.

Coplanar Vectors Vectors that lie in the same plane are said to be **coplanar**. We have just seen that if the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are not coplanar, then necessarily $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \neq 0$, since the volume of a parallelepiped with edges $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ has nonzero volume. Equivalently stated, this means that if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, then the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are coplanar. Since the converse of this last statement is also true, we have

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0 \text{ if and only if } \mathbf{a}, \mathbf{b}, \text{ and } \mathbf{c} \text{ are coplanar.}$$

Physical Interpretation of the Cross Product In physics a force $\mathbf{F}$ acting at the end of a position vector $\mathbf{r}$, as shown in **FIGURE 7.4.5**, is said to produce a **torque** $\boldsymbol{\tau}$ defined by $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$. For example, if $\|\mathbf{F}\| = 20 \text{ N}$, $\|\mathbf{r}\| = 3.5 \text{ m}$, and $\theta = 30^\circ$, then from (11),

$$\|\boldsymbol{\tau}\| = (3.5)(20) \sin 30^\circ = 35 \text{ N}\cdot\text{m}.$$

If $\mathbf{F}$ and $\mathbf{r}$ are in the plane of the page, the right-hand rule implies that the direction of $\boldsymbol{\tau}$ is outward from, and perpendicular to, the page (toward the reader).

As we see in **FIGURE 7.4.6**, when a force $\mathbf{F}$ is applied to a wrench, the magnitude of the torque $\boldsymbol{\tau}$ is a measure of the turning effect about the pivot point P and the vector $\boldsymbol{\tau}$ is directed along the axis of the bolt. In this case $\boldsymbol{\tau}$ points inward from the page.

Remarks

When working with vectors, one should be careful not to mix the symbols $\cdot$ and $\times$ with the symbols for ordinary multiplication, and to be especially careful in the use, or lack of use, of parentheses. For example, expressions such as

$$\mathbf{a} \times \mathbf{b} \times \mathbf{c} \quad \mathbf{a} \cdot \mathbf{b} \times \mathbf{c} \quad \mathbf{a} \cdot \mathbf{b} \cdot \mathbf{c} \quad \mathbf{a} \cdot \mathbf{bc}$$

are not meaningful or well-defined.

7.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-15.

In Problems 1–10, find $\mathbf{a} \times \mathbf{b}$.

- $\mathbf{a} = \mathbf{i} - \mathbf{j}$, $\mathbf{b} = 3\mathbf{j} + 5\mathbf{k}$
- $\mathbf{a} = 2\mathbf{i} + \mathbf{j}$, $\mathbf{b} = 4\mathbf{i} - \mathbf{k}$
- $\mathbf{a} = \langle 1, -3, 1 \rangle$, $\mathbf{b} = \langle 2, 0, 4 \rangle$
- $\mathbf{a} = \langle 1, 1, 1 \rangle$, $\mathbf{b} = \langle -5, 2, 3 \rangle$
- $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{b} = -\mathbf{i} + 3\mathbf{j} - \mathbf{k}$
- $\mathbf{a} = 4\mathbf{i} + \mathbf{j} - 5\mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$
- $\mathbf{a} = \langle \frac{1}{2}, 0, \frac{1}{2} \rangle$, $\mathbf{b} = \langle 4, 6, 0 \rangle$
- $\mathbf{a} = \langle 0, 5, 0 \rangle$, $\mathbf{b} = \langle 2, -3, 4 \rangle$
- $\mathbf{a} = \langle 2, 2, -4 \rangle$, $\mathbf{b} = \langle -3, -3, 6 \rangle$
- $\mathbf{a} = \langle 8, 1, -6 \rangle$, $\mathbf{b} = \langle 1, -2, 10 \rangle$

In Problems 11 and 12, find $\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3}$.

- $P_1(2, 1, 3)$, $P_2(0, 3, -1)$, $P_3(-1, 2, 4)$
- $P_1(0, 0, 1)$, $P_2(0, 1, 2)$, $P_3(1, 2, 3)$

In Problems 13 and 14, find a vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

- $\mathbf{a} = 2\mathbf{i} + 7\mathbf{j} - 4\mathbf{k}$, $\mathbf{b} = \mathbf{i} + \mathbf{j} - \mathbf{k}$
- $\mathbf{a} = \langle -1, -2, 4 \rangle$, $\mathbf{b} = \langle 4, -1, 0 \rangle$

In Problems 15 and 16, verify that $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$ and $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

- $\mathbf{a} = \langle 5, -2, 1 \rangle$, $\mathbf{b} = \langle 2, 0, -7 \rangle$